ZOO501-Current Mid Terms 2022

COMPILED BY MUHAMMAD KASHIF

1)Differnce b/w Karyokinesis and Cytokinesis

Karyokinesis the mitotic division of the nucleus.

The mechanical agent of this division is the mitotic spindle, with its microtubules composed

of tubulin (the same type of protein that makes up the sperm flagellum).

The second process is cytokinesis the division of the cell.

The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin (the same type of protein that extends the egg microvilli and the sperm acrosomal process).

The relationship and coordination between these two systems during cleavage, in which a sea urchin egg is shown undergoing first cleavage.

The mitotic spindle and contractile ring are perpendicular to each other, and the spindle is internal to the contractile ring. The contractile ring creates a cleavage furrow, which eventually bisects the plane of mitosis, thereby creating two genetically equivalent blastomeres.

2)Polyadenylation

is the addition of a poly(A) tail to an RNA transcript, typically a messenger RNA (mRNA). The poly(A) tail consists of multiple adenosine monophosphates; in other words, it is a stretch of RNA that has only adenine bases. ... In many bacteria, the poly(A) tail promotes degradation of the mRNA.

The poly(A) tail

- Confers stability on the messenger RNA.
- Allows the mRNA to exit the nucleus.
- Permits the mRNA to be translated into protein.

3)Marginal Zone/Belt

The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area opaca.Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone (or marginal belt).**

4)Area Pellucida

Between the blastoderm and the yolk is a space called the subgerminal cavity. This space is created when the blastoderm cells

absorb fluid from the albumin ("egg white") and secrete it between themselves and the yolk. At this stage, the deep cells in the center of the blastoderm are shed and die, leaving behind a onecell- thick area pellucida.

5)Area Opaca

The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the **area opaca**

6)Dendrites

The fine extensions of the neuron that are used to pick up electrical impulses from other cells

are called dendrites. Some neurons develop only a few dendrites, whereas other cells (such as the Purkinje neurons) develop extensive dendritic trees.

7)Fertilization:

The process of combining the male gamete, or sperm, with the female gamete, or ovum. The product of fertilization is a cell called a zygote.

8)Somites

Once the neural tube has formed, it induces changes in its neighbors, and organogenesis continues. The mesodermal tissue adjacent to the notochord becomes segmented into somites, the precursors of

the frog's back muscles, spinal cord, and dermis (the inner portion of the skin).

9)Tadpole

A tadpole is the larval stage in the life cycle of an amphibian. Most tadpoles are fully aquatic, though some species of amphibians have tadpoles that are terrestrial. Tadpoles have some features that may not be found in adult amphibians such as a lateral line, gills, and tails.Somites appear as blocks of mesodermal tissue. The embryo develops a mouth and an anus, and it elongates into the typical tadpole structure.

10)Neural Tube

A rod of mesodermal cells in the most dorsal portion of the embryo. At this stage, the embryo is called a Neurula. The neural precursor cells elongate, stretch, and fold into the embryo forming the neural tube.

11)Differences between Primary & Secondary Neurulation?

There are two major ways of forming a neural tube

- **Primary neurulation**
- **Secondary neurulation**

- In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.
- In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.
- In amphibians, such as Xenopus, most of the tadpole neural tube is made by primary neurulation, but the tail neural tube is derived from secondary neurulation.
- In mice (and probably humans, too), secondary neurulation begins at or around the level of somite 35.

12)Interaction between Egg & Sperm Cells in Simple Steps

The interaction of sperm and egg generally proceeds according to five basic step:

- The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
- The exocytosis of the acrosomal vesicle to release its enzymes.
- The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
- The passing of the sperm through this extracellular envelope.
- Fusion of egg and sperm cell plasma membranes.

13)Two Main Strategies of Developmental Commitment ?

The interaction of sperm and egg generally proceeds according to five basic step:

- ➢ The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.
- ➢ The exocytosis of the acrosomal vesicle to release its enzymes.
- ➢ The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.
- ➢ The passing of the sperm through this extracellular envelope.
- ➢ Fusion of egg and sperm cell plasma membranes.

The perception of external inductive signals: The process where one cell or group of cells

changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

- ▪ Two identical cells can follow alternative fates if one is exposed to an external signal (often secreted by a different cell).

14)MHP

In birds and mammals, the cells at the midline of the neural plate are called the medial hinge point (MHP) cells.

15)Role of Calcium(Ca+) in Fusion of Sperm & Egg.

Role of calcium in cortical reaction

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

16)Transcription Factor !

Transcription factors

Transcription factors are proteins that bind to enhancer or promoter regions and interact to activate or repress the transcription of a particular gene. Most transcription factors can bind to specific DNA sequences. These proteins can be grouped together in families based on similarities in structure.

17)Gene Regulation

Gene regulation is the process of turning genes on and off. During early development, cells begin to take on specific functions. Gene regulation ensures that the appropriate genes are expressed at the proper times. Gene regulation can also help an organism respond to its environment.

18)How to Gain Gene Activity knowledge in humans

Knowledge of gene activity in humans can be obtained by candidate gene mapping or positional cloning.

19)PreChordal Plate

In the development of vertebrate animals, the prechordal plate is a "uniquely thickened portion" of the endoderm that is in contact with ectoderm immediately rostral to the cephalic tip of the notochord. It is the most likely origin of the rostral cranial mesoderm.

The prechordal plate induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

20)Chordamesoderm

Axial mesoderm, or chordamesoderm, is the mesoderm in the embryo that lies along the central axis under the neural tube. will give rise to notochord starts as the notochordal process, whose formation finishes at day 20 in humans.the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord. The first cells migrating laterally through the primitive streak become endoderm, displacing the hypoblast.

21)Transcription factors domains

Transcription factors have **three major domains.**

1) DNA-binding domain

2) Protein-protein interaction domain

3) Trans-activating domain ➢ DNA-binding domain:

It recognizes a particular DNA sequence.

➢ Protein-protein interaction domain:

It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.

➢ Trans-activating domain:

It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this trans-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.

22)Recognition Steps Between Sperm and Egg

Recognition between sperm and egg involves

molecules on the acrosomal process. Together, the acrosome and nucleus constitute the head of the sperm.

The means by which sperm are propelled vary according to how the species has adapted to environmental conditions. In some species (such as the parasitic roundworm Ascaris), the sperm travel by the amoeboid motion of lamellipodial extensions of the cell membrane.

In most species, however, each sperm is able to travel long distances by whipping its flagellum. Flagella are complex structures. The major motor portion of the flagellum is called the axoneme. It is formed by microtubules emanating from the centriole at the base of the sperm nucleus.

The core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules.The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

The sperm released during ejaculation are able to move, yet they do not have the capacity to bind to and fertilize an egg. These final stages of sperm maturation (called capacitation) do not occur until the sperm has been inside the female reproductive tract for a certain period of time.

23)Syncytium

In early embryos of these insects, cell division is not complete. Rather, the nuclei divide within the egg cytoplasm. This creates many nuclei in the large egg cell.

A cytoplasm that contains many nuclei is called a syncytium.

24)Why Ascidians are Called Tunicates

As larvae, Ascidians are free-swimming tadpoles; but when the tadpole undergoes metamorphosis, it sticks to the sea floor, its nerve cord and notochord degenerate, and it secrete a cellulose yunic (which gave the name "tunicates" to these creatures).Thats why , they are called as Tunicates.

25)Steps in Gene Regulation

The regulation of gene expression can be accomplished at several levels:

- **Differential gene transcription**

It regulates that which of the nuclear genes are transcribed into RNA.

- **Selective nuclear RNA processing:**

It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.

- **Selective messenger RNA translation:**

It regulates that which of the mRNAs in the cytoplasm become translated into proteins.

- **Differential protein modification:**

It regulates that which proteins are allowed to remain or function in the cell.

26)Resact

The mechanisms of chemotaxis differ among species. One chemotactic molecule, a 14-amino acid peptide called resact, has been isolated from the egg jelly of the sea urchin Arbacia punctulata.

27)Life Cycle of Chlamydomonas

The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These

specialized regions contain mating type-specific components that enable the cytoplasms to fuse.

Following flagellar agglutination, the plus individuals initiate fusion by extending a fertilization tube. This tube contacts and fuses with a specific site on the minus individual.

In evolving sexual reproduction, two important advances had to be achieved.

➢ The first was the mechanism of meiosis whereby the diploid complement of chromosomes is reduced to the haploid state.

➢ The second was a mechanism whereby the two different mating types could recognize each other.

28)Fusogenic & Fertilin Protein

Fusion is an active process, often mediated by specific "fusogenic" proteins.

Gamete Fusion in Mammals

In mammals, the fertilin proteins in the sperm plasma membrane are essential for sperm membrane-egg membrane fusion. Mouse fertilin is localized to the posterior plasma membrane of the sperm head. It adheres the sperm to the egg by binding to the 6β1 integrin protein on the egg plasma membrane. Moreover, like sea urchin bindin (to which it is not structurally related), fertilin has a hydrophobic region that could potentially mediate the union of the two membranes.

29)MPF Sub-Units

Mitosis-promoting factor **(MPF) contains two subunits.**

- **The large subunit is called cyclin B.**
- **Small subunit of MPF, the cyclin-dependent kinase**
- Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

30)Gastrulation in Snails

The snail stereoblastula is relatively small.

Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres. Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole.

In the typical cleavage of molluscs, cells divides to produce most of the larval structures, including a gland capable of producing a large shell.

31)Silencers

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

32)Deuterostomes:

 In Deuterostomes ("mouth second"), the mouth opening is formed after the anal opening.

 Phyla in the Deuterostomes lineage include the chordates and echinoderms.

33)Early & Late Response in Egg & Sperm

Early responses:

Contact between sea urchin sperm and egg activates the two major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, and the slow block, initiated by the

intracellular release of calcium ions. The activation of all eggs appears to depend on an increase in the concentration of free calcium ions within the egg. Such an increase can occur in two ways:

calcium ions can enter the egg from outside, or calcium ions can be released from the endoplasmic reticulum within the egg.

Late responses:

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases.

The rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg.

This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

34)PROTOSTOMES:

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

35)Exon and Intron Difference

Exons:

Coding regions of DNA.

• **Introns**:

Non-coding regions.

36)Gray & White Matter

Glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

37)Development Strategies

The inheritance of cytoplasmic determinants: Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic

determinants indicates that the mechanism of differentiation is entirely intrinsic.

The perception of external inductive signals:

The process where one cell or group of cells

changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

- Two identical cells can follow alternative fates if one is exposed to an external signal (often secreted by a different cell).

38)Blastocyst

A blastomere is a cell derived from cleavage in an early embryo. A blastula is an embryonic structure composed of blastomeres. The cavity in the blastula is the blastocoel. A mammalian blastula is called a blastocyst.

39)Capacitation

The sperm released during ejaculation are able to move, yet they do not have the capacity to bind to and fertilize an egg. These final stages of sperm maturation (called capacitation) do not occur until the sperm has been inside the female reproductive tract for a certain period of time.

40)What causes this cyclic activity of MPF?

Mitosis-promoting factor contains two subunits.

- **The large subunit is called cyclin B.**
- **Small subunit of MPF, the cyclin-dependent kinase**
- **Cyclin B is often encoded by mRNAs stored in the oocyte** cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

41)Function of Estrogen & Progesterone

Estrogen is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg. Another ovarian hormone, progesterone, signals the egg to resume its meiotic division. This is necessary because the egg had been "frozen" in the metaphase of its first meiosis. When it has completed this first meiotic division, the egg is released from the ovary and can be fertilized.

42)Birds(Anterior _Posterior Axis Early Development fate)

In birds, gravity is critical in determining the anterior-posterior axis, while pH differences appear crucial for distinguishing dorsal from ventral. The left-right axis is formed by the expression of nodal on the left side of the embryo.

43)Potency and cell fate

1. **Potency** is the entire range of the cell types a particular cell can give rise to in all possible environments.

2. For example: A cell is pluripotent if it can give rise to all different blood cell lineages. There is a decrease in potency as cells become committed.

Fate of a cell is all the different cell types, its descendants can become during normal development.

- ➢ **Cell potency ≥ fate.**
- ➢ **Potency is intrinsic property.**
- ➢ **Cell fate is equal to potency plus environment.**
- ➢ **Cell fate becomes increasingly restricted until a cell is terminally differentiated.**

44)Functional Domains of Neural Crest

These regions of neural crest can be divided into four main functional domains, which include the cranial neural crest, trunk neural crest, vagal and sacral neural crest, and cardiac neuraspecification

45)MODES OF COMMITMENT

Three basic modes of commitment are,

- ❖ **Autonomous specification**
- ❖ **Conditional specification**
- ❖ **Syncytial specification**

46)Neural Tube Function in Tunicates

The neural tube is the embryonic structure that ultimately forms the brain and spinal cord. The neural tube is formed during a process called neurulation, which begins when the underlying dorsal mesoderm signals the ectodermal cells above it to elongate into columnar neural plate cells

47)Zona Pellucida_Cumulus_Corona Radiata

The vitelline envelope is essential for the species-specific binding of sperm. In mammals, the vitelline envelope is a separate and thick extracellular matrix called the zona pellucid.

The mammalian egg is also surrounded by a layer of cells called the cumulus.

The innermost layer of cumulus cells, immediately adjacent to the zona pellucida, is called the corona radiata.

48)Cortical Granules

Cortical granules are membrane-bound structures containing proteolytic enzymes. However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules.

49)Periderm

The second cell population distinguished at the midblastula transition is the enveloping layer (EVL).It is made up of the most superficial cells of the blastoderm, which form an epithelial sheet a

single cell layer thick. The EVL eventually becomes the periderm, an extra-embryonic protective covering that is sloughed off during later development.

50)Difference between Axoneme & Dynein

The major motor portion of the flagellum is called the axoneme. It is formed by microtubules emanating from the centriole at the base of the sperm nucleus.

The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

51)Ways in Which Gene Can influence Activity of Other Genes

Over the past decade, six generalizations that emphasize their importance for differential gene expression have been made:

 Most genes require enhancers for their transcription.

 Enhancers are the major determinant of differential transcription in space (cell type) and time.

 A given gene can have several enhancer sites linked to it, and each enhancer can be bound by more than one transcription factor.

 The interaction between the proteins bound to the enhancer sites and the transcription initiation complex assembled at the promoter is thought to regulate transcription.

 Enhancers are modular. It is the combination of transcription factors that causes particular genes to be transcribed.

 A gene can have several enhancer elements, each turning it on in a different set of cells.

 Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. Only small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.
21. The division of embryo into 3 Germ layers
22. Cell potency ___≥___fate
23. Zona pellucida also initiate the acrosomal reaction after sperm bound to it.
24. Mammalian egg is surrounded by layer called. Cumulus
25. The volume of the sea urchin is about 200 Picoliters.
26. The sperm travel by the amoeboid motion of lamellipodial extensions of the cell membrane.
27. The sex and Reproduction process included into fertilization.
28. The term spermatozoa means.. Sperm Animals
29. In 1824 J. L. Prevost and J. B. Dumasclaimed sperm is not parasite.

Join VU BIG ANIMAL

30. 4 cadherin classes are found in Vertebrates embryos. .
31. The second major group of protostomes are the Lophotrochozoa
32. Sperm discovered in 1678
33. Transcription factors have three major domains
34. How many Amphibian oocytes produces ribosomes as 10^{12}
35. Enhancers can also be used to inhibit transcription
36. In deutrosomes the mouth opening would formed after... opening Anal
37. Germs cell formed in.... specialized portion of egg cytoplasm
38. Syncytial specification found in insects
39. Third anatomical approach to developmental biology is Teratology
40. The mature sea urchin is metabolically sluggish and activated by. Sperm
41. The resting membrane potential is.....mv. 70mV
42. Fusion is a Active process that is activated by fusogenic proteins.
43. Axoneme s formed by. Microtubules
44. Acrosome can be modified secreted by..... Golgi apparatus (not sure)
45. The male leopard frog release sperm in the summer.
46. Eukaryotic genes are contained within a complex of DNA and protein called Chromatin
47. The negative enhancers are called Silencers
48. The ability of embryonic cells to change their fates is called..... Regulation
49. Pax6 transcription factor found in mammalian eye, nervous system and pancreas.
50. E- Cadherin is expressed on all early mammalian embryonic cells.
51. Annelids, molluscs and flatworms belong to..... Lophotrochozoa
52. Potency of cell decrease if cell is committed.
53. Fertilization in Rana Pipiens usually lasts for about 3 (three) years.
54. An uncommitted cell can be described as Naive
55. Lying immediately beneath the plasma membrane of the egg is a thin Shell
56. Acroomal vesicles are considered as modified secretory vesicles
57. Most marine organisms release their gametes into the environment
58. The fast block method is used to prevent Polyspermy
59. Autonomous specification was first demonstrated in 1887
60. In mouse ZP3 is attached to zona pellucid
61. TAFs used in transcription factor activity
62. in slow block method which is cortical granule protein hyalin and peroxidase
63. Egg shell formed weight of egg. 9 – 12%
64. potency is intrinsic property
65. cytoplasm that contains many nuclei called syncytium
66. Ectoderm forms..... epidermis, nervous system, and pigment cells.
67. according to dutch sperm are ... parasitic animals (spermatozoa)
68. Acrosome & nucleus constitute the head of the sperm.
69. cortical granules are membrane bound structures
70. Acrosomal reaction has 2 (two) component
71. Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules
72. Ca & bicarbonate ions maybe critical in activating cAMP

Join VU BIO ANIMALS

73. Peroxidase enzyme hardens fertilization envelope
74. The Fast block to polyspermy is achieved by changing the electric potential
75. P-cadherin stands for _____Placental Cadherin
76. Most transcription factor can bind to specific DNA sequences
77. MITF protein necessary for production of pigments cell and their pigments.
78. Outside of plasma membrane is the vitelline envelope.
79. Mouse __fertilin___is localized to posterior plasma membrane of the sperm.
80. Finally the fourth cortical granules protein ___hyaline___ forms a coating around egg.
81. The cleavage stages are called.____Blastomere
82. Enhancers are....._DNA Sequences
83. The mitotic spindle and contractile ring are perpendicular to each other
84. J.Whittaker stained blastomeres for the presence of the enzyme acetyl cholinesterase
85. Protocadherins are involved in separating notochord from mesoderm in __Xenopus gastrulation
86. Pax6 factor needed for mammalian eye
87. Which stands comparative embryology is ____First_
88. Cell type specification precedes any large-scale embryonic cell migration
89. Series of extremely rapid divisions in zygote is ___Cleavage
90. In protosome, mouth is formed first than Deuterostomes: forms.
91. Northern blots, in situ hybridization, & the PCR can show which cells are transcribing particular genes.
92. Coding regions of DNA are called __Exon
93. Na and __K__ Are important in fast block polyspermy.
94. The force for sperm propulsion is provided by __dynein
95. Spallenazani demonstrated that filtered __toad__ semen devoid of sperm would not fertilize egg.
96. Mesoderm generates the____KIDNEY.
97. Sperms are attracted towards egg of their species by CHEMOTAXIS.
98. Mathematical modeling seeks to desire developmental phenomenon in terms of EQUATIONS.
99. A region that is not translated in proteins is called UNTRANSLATED region.
100. Invariable cleavage produce the ___SAME___ lineages in embryo
101. Germ layers are formed as a result of ___GASTRULATION
102. RNA polymerase binds with PROMOTOR in start of transcription
103. The region of globular actin molecule lies between nuclear and acrosomal vesicle in ____SEA URCHIN
104. Which technique is used in situ hybridization ____Northern blots
105. Conditional specification involves interaction with RELATIVE POSITIONS cells
106. Particular proteins or carbohydrate on the sperm surface __LOST__ during capacitation
107. ___LOSS____of hydrogen ion causes pH to rise.
108. Peroxidase enzyme hardens the fertilization envelope by crosslinking TYROSINE residues
109. FERTILIZATION transmits genes from parents to offspring.
110. Regulation of gene expression accomplished by SELECTIVE MESSENGER RNA TRANSLATION
111. Development contact genotype and PHENOTYPE
112. Nucleosome is composed of OCTAMER of histone protein.
113. Knowledge of gene activity in humans can be obtained by candidate GENE MAPPING & POSITIONAL CLONING
114. Anatomical abnormalities are caused by MUTANT GENES
115. All multicellular organisms arise by a slow process of progressive change called DEVELOPMNET

Join VU BIO ANIMALS

116. Animals development is the division of embryo into three layers Ectoderm, Mesoderm and ENDODERM
117. The ions which are responsible for increasing granule reactions Ca (Calcium)
118. The zona pellucida is a fglycoprotein matrix, which is synthesized and secreted by the growing oocyte
119. In several other species, certain regions of the membrane are specialized for sperm recognition & fusion
120. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development
121. Activation of egg metabolism to start development
122. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates
123. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in Acetabularia.
124. Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly
125. Bacteria are able to transmit genes from one individual to another by means of sex pili
126. +ve and – ve mating take place *Chlamydomonas*

Join VU BIO ANIMALS

ZOO501 FINAL TERM MCQS

1. Mode of commitment_____3___
2. strageries of commitment_____2___
3. Rana pipiens usually lays around __2500 eggs_____while the bullfrog, Rana catesbiana, can lay as many as 20,000
4. Multinucleated type of cell form._____syncytiotrophoblast
5. slow block, initiated by the intracellular release of_____ calcium ions
6. There are_____ two ____ major modes of bone formation, or **osteogenesis**
7. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body
8. Human brain contains_____**10^{12}** glial cells____
9. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
10. ____Pax6 _____ plays a major role in eye formation
11. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
12. The eggs of insects have their yolk in the ____center_____ (i.e., they are centrolecithal),
13. Egg activates due to which ion more _____Ca____
14. Which dermis give rise to brain _____Ectoderm____
15. Which dermis give rise to eye _____Ectoderm_____
16. Mod of commitment in insects _____syncytial specification____
17. Regulative development arises from_____ conditional specification
18. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___ of DNA
19. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
20. Paracrine factors such as TGF- and FG**F7** are important in normal _____skin development
21. The male leopard ___frog___release sperm in the summer
22. In mouse ZP3 is attached to_____zona pellucid
23. It is estimate ____egg is 25000 to 50000 mRNA. _____Sea Urchin
24. Fertilin protein is important for sperm-egg fusion in _____Mammals___
25. Fast block to polyspermy is initiated by _____Sodium___
26. Transition from fertilization to cleavage is caused by factor _____MPF
27. Eggs having sparse and equally yolk space are called _____isolethial___
28. Archenteron is used for _____primitive gut___
29. Spiral cleavage found in _____Snails____
30. Sonic hedgehog is a type of _____protein____
31. Which type of neurulation in fish _____secondary___
32. Nural crest divides functionally in domains _____4_____
33. Myogenic bHLH proteins are specific for development of _____muscle cell___

34. Zygote which divide mitotically to produce all the cells of the body.
35. Tatratology the study of birth defect.
36. Identical twins occur in roughly 0.25% of human births.
37. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.
38. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.
39. The speed of metamorphosis is directly linked with environment pressures
40. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
41. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.
42. 140 base pairs of DNA nucleosome.
43. Viteline envelop contains at least eight different glycoproteins.
44. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani
45. The volume of the sea urchin is about 200 Pico liters
46. Hindbrain Rhombencephalon
47. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction
48. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
49. The neural form keel fish
50. Spiral holoblastic clevage is characteristic of several animals groups, including annelid worms, some flateworms, and most mollusks(snails).
51. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm (hydra)
52. Cell fate is determined by signalling. The micromeres constitute a major signalling center. B-catenin is important for the inducing capacity of micromeres.
53. Even distribution of yolk in ovum isolecithal
54. Size of bindin protein 30,500-Da
55. which has two parts in cell cycle options Blastomere
56. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**
57. Which forms spinal cord, hind brain and mid brain chordamesoderm.
58. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
59. I will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the Decidua.
60. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
61. The mitotic division of the nucleus is called karyokinesis
62. Fishes, reptiles , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
63. The fast block, initiated by sodium influx into the cell in sea urchin.

64. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.

65. The rudiment of the central nervous system, is called neurulation.

66. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.

67. Fraternal twins are the result of two separate fertilization events.

68. The prechordal ____plate_____induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

69. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.

ZOO501 CURRENT FINAL TERM MCQS

1. Due to **Loss** in hydrogen ions, pH rises.
2. As a result of gastrulation, ectoderm, mesoderm and **Endoderm** are formed.
3. Male leopard **Frogs** make their sperm in summer.
4. **Differential gene transcription** is step for gene expression regulation.
5. Coding regions of DNA **exon.**
6. How many domains are present for transcription **3**
7. pdx1 is present in **Pancreatic** And Duodenal1 gene.
8. Bacteria transmit genes from one cell to other by **sex pili.**
9. Cells rearrange their **DNA** to create new immunoglobin and antigen receptor gene.
10. Sea urchins contain **25,000 to 50,000** mRNA.
11. In many species, sperm are attracted towards eggs by **chemotaxis**.
12. Acrosomal protein mediating recognition by protein **bindin.**
13. Sea water has **High** sodium ion conc.
14. Eggs have sparse **equally** spaced yolk are icolecithel.
15. Cytoplasm of blastodisc become embryo this meroblastic division called **discoidal.**
16. **The micromere** constitute major signaling center.
17. Osteoblasts secrete collagen proteoglycan **matrix** that is able to bind calcium salts
18. Tgf beta family secreted from the surface **ectoderm** and roof of neural tube.
19. Dendrites are numerous and do not extend far from the neuronal cell body or **soma,** axons may extend for several feet.
20. One agent of ventral specification is **sonic** hedgegog protein.
21. In frogs and chicks, **secondary** nueralation is seen in neural tube.
22. The cells at dorsal most portion of neural tube become **neural crest cells.**
23. About **33%** of identical twins have two complete and separate chorions
24. Mammals undergo holoblastic **rotational** cleavage.
25. In birds, **gravity** is critical in determining anterior posterior axis
26. Decidua becomes rich in blood vessels that will provide **oxygen** and nutrients to embryo.
27. Mammalian oocyte is released from ovary and swept by fimbriae into **the oviduct.**
28. Fertilization in chick egg occurs in **oviduct.**
29. Developmental decisions are usually **irreversible.**
30. All multicellular organisms arise by a slow process of progressive change called **development**.
31. Major class of cadherin includes **N- cadherin.**
32. Only **small %** of genome is expressed in partial cell.
33. Cortical are **granules** are membrane bound structure containing proteolytic enzymes.
34. Small subunits of MPF **cyclin-dependent kinase.**
35. It secretes a **cellulose** tunic (which gave the name "tunicates" to these creatures).
36. Inner surface cells thin 30×5um process **filopodia.**
37. β-catenin is Important for Inducing capacity of **micromeres.**
38. In fishes , First **12** divisions occur synchronously form a mouln of cells that sits an animal pole of large yolk cell.
39. Fertilization occurs in **ampulla** of oviduct a close, a region close to ovary.
40. In amphibians most of the xenopus most of tadpole is formed by **primary neurulation** and tail of neural tube formed by 2ndry neuralation.
41. Neural fold migrates to midline of embryo and fusing to form **neural tube** beneath the overlying ectoderm.

Join VU BIOANIMALS

42. **Tissues** form nervous system in the fly or in the frog.
43. The marginal zones to form clusters of neurons called **nuclei.**
44. Paracrine factors instruct **Myotome cells** to become muscles by inducing them to synthesize MyoD protein.
45. Sclerix is expressed in mesenchyme **from** the scleretome.
46. Acrosomal reaction in **sea urchins** is often highly specific to each species.
47. Cells are arrange their.. to creat new immunoglobin
48. In frogs and chicks, **secondary neurulation** is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.
49. Regulation of gene expression by **several level.**
50. Which type of neuralation in fish **exclusively secondary**.
51. sonic hedgehocy is a type of **protein.**
52. **Blastomeres** have two parts of cell cycle.
53. Number of cortical granules Sea urchin **15000.**
54. How many strategies involve in cell commitment **2.**
55. Resting membrane potential of the egg is **70mv.**
56. One chemotactic molecule, a 14-amino acid peptide called **resact**.
57. Force for sperm propulsion is provided by **dynin.**
58. Regulative development arises from **Conditional Specification.**
59. Human brain contains **10^{12} glial cells.**
60. Fish have discoidal **telolicithal** pattern.
61. Process of osteogensis **2**
62. The mitotic spindle and contractile ring are **perpendicular** to each other.
63. Gastrulation in Snails is accomplished primarily by **epiboly.**
64. The first cell movement of fish gastrulation is the **epiboly** of the blastoderm.
65. This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells.
66. Secondary neurulation involves the making of a **medullary cord** and its subsequent hollowing into neural tube
67. The mantle zone cells differentiate into both **neurons** and **glia.**
68. The cerebral cortex in humans has **six** layers.
69. Paracrine factors such as **TGF** and FGF7 are important in normal skin development.
70. The **fifth** phase involves the invasion of the cartilage model by blood vessels.

71. Mode of commitment_____3___
72. strageries of commitment_____2___
73. Rana pipiens usually lays around __2500 eggs_____while the bullfrog, Rana catesbiana, can lay as many as 20,000
74. Multinucleated type of cell form._____syncytiotrophoblast
75. slow block, initiated by the intracellular release of_____ calcium ions
76. There are_____two____ major modes of bone formation, or **osteogenesis**
77. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia,
78. Human brain contains____**10^{12}** glial cells____
79. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
80. ____Pax6 _____ plays a major role in eye formation
81. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
82. The eggs of insects have their yolk in the ____center____ (i.e., they are centrolecithal),
83. Egg activates due to which ion more ____Ca____
84. Which dermis give rise to brain _____Ectoderm____
85. Which dermis give rise to eye _____Ectoderm_____
86. Mod of commitment in insects _____syncytial specification____
87. Regulative development arises from_____ conditional specification
88. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___of DNA
89. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
90. Paracrine factors such as TGF- and FG**F7** are important in normal _____skin development
91. In mouse ZP3 is attached to_____zona pellucid

Join VU BIO

92. It is estimate ____egg is 25000 to 50000 mRNA.
Sea Urchin
93. Fertilin protein is important for sperm-egg fusion in Mammals
94. Fast block to polyspermy is initiated by Sodium
95. Transition from fertilization to cleavage is caused by factor MPF
96. Eggs having sparse and equally yolk space are called isolethial
97. Archenteron is used for primitive gut
98. Spiral cleavage found in Snails
99. Sonic hedgehog is a type of protein

1. Which type of neurulation in fish secondary
2. Nural crest divides functionally in domains 4
3. Myogenic bHLH proteins are specific for development of muscle cell
4. Zygote which divide mitotically to produce all the cells of the body.
5. Tatratology the study of birth defect.
6. Identical twins occur in roughly 0.25 of human births.
7. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.
8. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.
9. The speed of metamorphosis is directly linked with environment pressures
10. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
11. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.
12. 140 base pairs of DNA nucleosome.
13. Viteline envelop contains at least eight different glycoproteins.
14. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani
15. The volume of the sea urchin is about 200 Pico liters
16. Hindbrain Rhombencephalon
17. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction
18. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
19. The neural form keel fish
20. Spiral holoblastic clevage is characteristic of several animals groups, including **annelid** worms, some flatworms, and most mollusks (snails).
21. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm (hydra)
22. Even distribution of yolk in ovum isolecithal
23. Size of bindin protein 30,500-Da
24. which has two parts in cell cycle options Blastomere
25. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**
26. Which forms spinal cord, hind brain and mid brain chordamesoderm.
27. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
28. It will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the Decidua.
29. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
30. The mitotic division of the nucleus is called karyokinesis
31. Fishes, reptiles , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
32. The fast block, initiated by sodium influx into the cell in sea urchin.
33. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.
34. The rudiment of the central nervous system, is called neurulation.
35. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.
36. Fraternal twins are the result of two separate fertilization events.
37. The prechordal plate induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

38. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.
39. D
40. Fertilization in the absence of sex occur through the __Binary Fission__
41. The macromeres divide meridionally, forming a tier of eight (8) cells
42. cells, derived from the micromeres, are called the primary mesenchyme.
43. Molluscus follow which type of cleavage ____spiral holoblastic cleavage
44. A mammalian blastula is called a blastocyst.
45. Which of the following undergo discoidal cleavage __fishes, reptiles and birds
46. The egg jelly factors that initiate the Acrosomal reaction in sea urchins are often highly specific to each species
47. The Metencephalon gives rise to the cerebellum
48. In primary neurulation, the original ectoderm is divided into three sets of cells
49. The growing region at the tip of the lip is called the chordoneural hinge
50. Neural Tube will form ___Brain and spinal cord
51. Each ICM (Inner Cell Mass) cell has the same potency
52. Variable cleavages produce no invariant fate assignments to cells
53. In early embryos of these insects, cell division is not complete
54. The neural crest cells originate at the dorsal most region of the neural tube
55. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs
56. the SOX9 gene, which encodes a DNAbinding protein
57. the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum
58. The second step is the alignment of the myoblasts together into chains
59. Cleavages sphere is known as __Blastula
60. Osteoclasts are probably derived from the same precursors as macrophage blood cells
61. charby desired to know the cause of birth defects
62. The direct conversion of mesenchymal tissue into bone is called intramembranous ossification
63. Those cells that remain integral components of the neural tube lining become ependymal cells
64. The original type of trophoblast cells constitute a layer called the cytotrophoblast
65. The major structural characteristic of avian, reptilian, & mammalian gastrulation is the primitive streak
66. The Blastomere cell cycle is governed by the synthesis and degradation of_____cyclin
67. Both have Anurans __and _. __Toad and frog
68. They have a notochord as larvae (chordates), but they lack vertebrae.
69. Acrosomal vesicles is derived from Golgi apparatus and contains __Enzymes__
70. The large subunit of MPF is _cyclin B
71. ZP3 is the specific glycoprotein in the mouse
72. MITF necessary for the production of __Pigment cells
73. Mutation of sox9 gene cause ____camptomelic dysplasia
74. Between the blastoderm and the yolk is a space called the sub-germinal cavity
75. The second step of gastrulation involves the involution of the mesoderm
76. The first cleavage furrow appears centrally in the blastodisc
77. Fertilization occurs in the ampulla of the oviduct
78. Mother provides fetus with __nutrients and oxygen
79. The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm)
80. The fetal portion of placenta formed by __Chorion__
81. Paracrine factor instruct myotome cells to become__ Muscles
82. When myoblasts stop dividing, secrete fibronectin

Join VU BIO ANIMALS

83. Retinoic Acid comes from **Somites**
84. Hox genes pattern the anterior-posterior axis and help to specify positions along that **axis**
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and **hypothalamic** brain regions
86. Endochondral ossification involves the formation of **cartilage**
87. J.Hammerling exchanged nuclei b/w two **morphologically** distinct species
88. When plus individuals initiate fusion by extending a fertilization tube. Which contacts & fuses with a specific site on the **minus** individual.
89. Promoter is **enhancer** which is thought to regulate transcription.
90. Identical twins share a common **Chorion**
91. In Birds, **pH** differences appear crucial for distinguishing dorsal from ventral
92. Which regulate function of Promoter Enhancer
93. The space between two neurons, where transmitter are released **synapse**
94. integral components of the neural tube lining become **ependymal**
95. Mantle cells differentiate into **Gray matter**
96. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area **opaca.**
97. During fertilization, the **actin** molecules fuse to form the long microfilaments.
98. A condition in which cell lack the instructions for development is **naive**
1. metamorphosis is initiated by hormones from the tadpole's **thyroid gland**
2. Brain has **100** billion neurons
3. Release of Ca from intracellular storage controlled by dye **Aequorin**
4. Nervous system develop by **Neural Tube**
5. Allows the mRNA to exit the nucleus **poly a tail**
6. Contact and ___between sperm and egg **Recognition**
7. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by ____in the late 1700s. **Lazzaro Spallanzani**
8. The early embryonic cells need a supply of energy and amino acids which is provided by **yolk** proteins in the egg.
9. The depression is called the primitive **groove**

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. Only small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.

Join VU BIO ANIMALS

ZOO501 CURRENT FINAL TERM MCQS

1. Due to **Loss** in hydrogen ions, pH rises.
2. As a result of gastrulation, ectoderm, mesoderm and **Endoderm** are formed.
3. Male leopard **Frogs** make their sperm in summer.
4. **Differential gene transcription** is step for gene expression regulation.
5. Coding regions of DNA **exon.**
6. How many domains are present for transcription **3**
7. pdx1 is present in **Pancreatic** And Duodenal1 gene.
8. Bacteria transmit genes from one cell to other by **sex pili.**
9. Cells rearrange their **DNA** to create new immunoglobin and antigen receptor gene.
10. Sea urchins contain **25,000 to 50,000** mRNA.
11. In many species, sperm are attracted towards eggs by **chemotaxis**.
12. Acrosomal protein mediating recognition by protein **bindin.**
13. Sea water has **High** sodium ion conc.
14. Eggs have sparse **equally** spaced yolk are icolecithel.
15. Cytoplasm of blastodisc become embryo this meroblastic division called **discoidal.**
16. **The micromere** constitute major signaling center.
17. Osteoblasts secrete collagen proteoglycan **matrix** that is able to bind calcium salts
18. Tgf beta family secreted from the surface **ectoderm** and roof of neural tube.
19. Dendrites are numerous and do not extend far from the neuronal cell body or **soma,** axons may extend for several feet.
20. One agent of ventral specification is **sonic** hedgegog protein.
21. In frogs and chicks, **secondary** nueralation is seen in neural tube.
22. The cells at dorsal most portion of neural tube become **neural crest cells.**
23. About **33%** of identical twins have two complete and separate chorions
24. Mammals undergo holoblastic **rotational** cleavage.
25. In birds, **gravity** is critical in determining anterior posterior axis
26. Decidua becomes rich in blood vessels that will provide **oxygen** and nutrients to embryo.
27. Mammalian oocyte is released from ovary and swept by fimbriae into **the oviduct.**
28. Fertilization in chick egg occurs in **oviduct.**
29. Developmental decisions are usually **irreversible.**
30. All multicellular organisms arise by a slow process of progressive change called **development**.
31. Major class of cadherin includes **N- cadherin.**
32. Only **small %** of genome is expressed in partial cell.
33. Cortical are **granules** are membrane bound structure containing proteolytic enzymes.
34. Small subunits of MPF **cyclin-dependent kinase.**
35. It secretes a **cellulose** tunic (which gave the name "tunicates" to these creatures).
36. Inner surface cells thin 30×5um process **filopodia.**
37. β-catenin is Important for Inducing capacity of **micromeres.**
38. In fishes , First **12** divisions occur synchronously form a mouln of cells that sits an animal pole of large yolk cell.
39. Fertilization occurs in **ampulla** of oviduct a close, a region close to ovary.
40. In amphibians most of the xenopus most of tadpole is formed by **primary neurulation** and tail of neural tube formed by 2ndry neuralation.
41. Neural fold migrates to midline of embryo and fusing to form **neural tube** beneath the overlying ectoderm.

Join VU BIOANIMALS

42. **Tissues** form nervous system in the fly or in the frog.
43. The marginal zones to form clusters of neurons called **nuclei.**
44. Paracrine factors instruct **Myotome cells** to become muscles by inducing them to synthesize MyoD protein.
45. Sclerix is expressed in mesenchyme **from** the scleretome.
46. Acrosomal reaction in **sea urchins** is often highly specific to each species.
47. Cells are arrange their.. to creat new immunoglobin
48. In frogs and chicks, **secondary neurulation** is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.
49. Regulation of gene expression by **several level.**
50. Which type of neuralation in fish **exclusively secondary**.
51. sonic hedgehocy is a type of **protein.**
52. **Blastomeres** have two parts of cell cycle.
53. Number of cortical granules Sea urchin **15000.**
54. How many strategies involve in cell commitment **2.**
55. Resting membrane potential of the egg is **70mv.**
56. One chemotactic molecule, a 14-amino acid peptide called **resact**.
57. Force for sperm propulsion is provided by **dynin.**
58. Regulative development arises from **Conditional Specification.**
59. Human brain contains **10^{12} glial cells.**
60. Fish have discoidal **telolicithal** pattern.
61. Process of osteogensis **2**
62. The mitotic spindle and contractile ring are **perpendicular** to each other.
63. Gastrulation in Snails is accomplished primarily by **epiboly.**
64. The first cell movement of fish gastrulation is the **epiboly** of the blastoderm.
65. This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells.
66. Secondary neurulation involves the making of a **medullary cord** and its subsequent hollowing into neural tube
67. The mantle zone cells differentiate into both **neurons** and **glia.**
68. The cerebral cortex in humans has **six** layers.
69. Paracrine factors such as **TGF** and FGF7 are important in normal skin development.
70. The **fifth** phase involves the invasion of the cartilage model by blood vessels.

71. Mode of commitment_____3___
72. strageries of commitment_____2___
73. Rana pipiens usually lays around __2500 eggs_____while the bullfrog, Rana catesbiana, can lay as many as 20,000
74. Multinucleated type of cell form._____syncytiotrophoblast
75. slow block, initiated by the intracellular release of_____ calcium ions
76. There are_____two____ major modes of bone formation, or **osteogenesis**
77. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia,
78. Human brain contains____**10^{12}** glial cells____
79. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
80. ____Pax6 _____ plays a major role in eye formation
81. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
82. The eggs of insects have their yolk in the ____center____ (i.e., they are centrolecithal),
83. Egg activates due to which ion more ____Ca____
84. Which dermis give rise to brain _____Ectoderm____
85. Which dermis give rise to eye _____Ectoderm_____
86. Mod of commitment in insects _____syncytial specification____
87. Regulative development arises from_____ conditional specification
88. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___of DNA
89. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
90. Paracrine factors such as TGF- and FG**F7** are important in normal _____skin development
91. In mouse ZP3 is attached to_____zona pellucid

Join VU BIO

92. It is estimate ____egg is 25000 to 50000 mRNA.
Sea Urchin
93. Fertilin protein is important for sperm-egg fusion in Mammals
94. Fast block to polyspermy is initiated by Sodium
95. Transition from fertilization to cleavage is caused by factor MPF
96. Eggs having sparse and equally yolk space are called isolethial
97. Archenteron is used for primitive gut
98. Spiral cleavage found in Snails
99. Sonic hedgehog is a type of protein

1. Which type of neurulation in fish secondary
2. Nural crest divides functionally in domains 4
3. Myogenic bHLH proteins are specific for development of muscle cell
4. Zygote which divide mitotically to produce all the cells of the body.
5. Tatratology the study of birth defect.
6. Identical twins occur in roughly 0.25 of human births.
7. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.
8. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.
9. The speed of metamorphosis is directly linked with environment pressures
10. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
11. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.
12. 140 base pairs of DNA nucleosome.
13. Viteline envelop contains at least eight different glycoproteins.
14. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani
15. The volume of the sea urchin is about 200 Pico liters
16. Hindbrain Rhombencephalon
17. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction
18. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
19. The neural form keel fish
20. Spiral holoblastic clevage is characteristic of several animals groups, including **annelid** worms, some flatworms, and most mollusks (snails).
21. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm (hydra)
22. Even distribution of yolk in ovum isolecithal
23. Size of bindin protein 30,500-Da
24. which has two parts in cell cycle options Blastomere
25. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**
26. Which forms spinal cord, hind brain and mid brain chordamesoderm.
27. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
28. It will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the Decidua.
29. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
30. The mitotic division of the nucleus is called karyokinesis
31. Fishes, reptiles , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
32. The fast block, initiated by sodium influx into the cell in sea urchin.
33. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.
34. The rudiment of the central nervous system, is called neurulation.
35. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.
36. Fraternal twins are the result of two separate fertilization events.
37. The prechordal plate induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

38. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.
39. D
40. Fertilization in the absence of sex occur through the __Binary Fission__
41. The macromeres divide meridionally, forming a tier of eight (8) cells
42. cells, derived from the micromeres, are called the primary mesenchyme.
43. Molluscus follow which type of cleavage ____spiral holoblastic cleavage
44. A mammalian blastula is called a blastocyst.
45. Which of the following undergo discoidal cleavage __fishes, reptiles and birds
46. The egg jelly factors that initiate the Acrosomal reaction in sea urchins are often highly specific to each species
47. The Metencephalon gives rise to the cerebellum
48. In primary neurulation, the original ectoderm is divided into three sets of cells
49. The growing region at the tip of the lip is called the chordoneural hinge
50. Neural Tube will form ___Brain and spinal cord
51. Each ICM (Inner Cell Mass) cell has the same potency
52. Variable cleavages produce no invariant fate assignments to cells
53. In early embryos of these insects, cell division is not complete
54. The neural crest cells originate at the dorsal most region of the neural tube
55. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs
56. the SOX9 gene, which encodes a DNAbinding protein
57. the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum
58. The second step is the alignment of the myoblasts together into chains
59. Cleavages sphere is known as __Blastula
60. Osteoclasts are probably derived from the same precursors as macrophage blood cells
61. charby desired to know the cause of birth defects
62. The direct conversion of mesenchymal tissue into bone is called intramembranous ossification
63. Those cells that remain integral components of the neural tube lining become ependymal cells
64. The original type of trophoblast cells constitute a layer called the cytotrophoblast
65. The major structural characteristic of avian, reptilian, & mammalian gastrulation is the primitive streak
66. The Blastomere cell cycle is governed by the synthesis and degradation of_____cyclin
67. Both have Anurans __and _. __Toad and frog
68. They have a notochord as larvae (chordates), but they lack vertebrae.
69. Acrosomal vesicles is derived from Golgi apparatus and contains __Enzymes__
70. The large subunit of MPF is _cyclin B
71. ZP3 is the specific glycoprotein in the mouse
72. MITF necessary for the production of __Pigment cells
73. Mutation of sox9 gene cause ____camptomelic dysplasia
74. Between the blastoderm and the yolk is a space called the sub-germinal cavity
75. The second step of gastrulation involves the involution of the mesoderm
76. The first cleavage furrow appears centrally in the blastodisc
77. Fertilization occurs in the ampulla of the oviduct
78. Mother provides fetus with __nutrients and oxygen
79. The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm)
80. The fetal portion of placenta formed by __Chorion__
81. Paracrine factor instruct myotome cells to become__ Muscles
82. When myoblasts stop dividing, secrete fibronectin

Join VU BIO ANIMALS

83. Retinoic Acid comes from **Somites**
84. Hox genes pattern the anterior-posterior axis and help to specify positions along that **axis**
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and **hypothalamic** brain regions
86. Endochondral ossification involves the formation of **cartilage**
87. J.Hammerling exchanged nuclei b/w two **morphologically** distinct species
88. When plus individuals initiate fusion by extending a fertilization tube. Which contacts & fuses with a specific site on the **minus** individual.
89. Promoter is **enhancer** which is thought to regulate transcription.
90. Identical twins share a common **Chorion**
91. In Birds, **pH** differences appear crucial for distinguishing dorsal from ventral
92. Which regulate function of Promoter Enhancer
93. The space between two neurons, where transmitter are released **synapse**
94. integral components of the neural tube lining become **ependymal**
95. Mantle cells differentiate into **Gray matter**
96. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area **opaca.**
97. During fertilization, the **actin** molecules fuse to form the long microfilaments.
98. A condition in which cell lack the instructions for development is **naive**
1. metamorphosis is initiated by hormones from the tadpole's **thyroid gland**
2. Brain has **100** billion neurons
3. Release of Ca from intracellular storage controlled by dye **Aequorin**
4. Nervous system develop by **Neural Tube**
5. Allows the mRNA to exit the nucleus **poly a tail**
6. Contact and ___between sperm and egg **Recognition**
7. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by ____in the late 1700s. **Lazzaro Spallanzani**
8. The early embryonic cells need a supply of energy and amino acids which is provided by **yolk** proteins in the egg.
9. The depression is called the primitive **groove**

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. Only small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.

Join VU BIO ANIMALS

21. The division of embryo into 3 _____Germ layers
22. Cell potency___≥___fate
23. Zona pellucida_____ also initiate the acrosomal reaction after sperm bound to it.
24. Mammalian egg is surrounded by layer called. _____Cumulus
25. The volume of the sea urchin is about ___200____ Picoliters.
26. The sperm travel by the amoeboid motion of ____ lamellipodial extensions___of the cell membrane.
27. The sex and __Reproduction_____ process included into fertilization.
28. The term spermatozoa means.. _____Sperm Animals
29. In 1824____J. L. Prevost and J. B. Dumasclaimed__ sperm is not parasite.
30. 4 cadherin classes are found in _____Vertebrates____ embryos. .
31. The second major group of protostomes are the _____Lophotrochozoa__
32. Sperm discovered in_____1678__
33. Transcription factors have ___three___ major domains
34. How many Amphibian oocytes produces ribosomes as_____10^{12}
35. Enhancers can also be used to inhibit _____transcription___
36. In deutrosomes the mouth opening would formed after... opening _____Anal__
37. Germs cell formed in...._____specialized portion of egg cytoplasm
38. Syncytial specification found in ____insects___
39. Third anatomical approach to developmental biology is ____Teratology___
40. The mature sea urchin is metabolically sluggish and activated by. _____Sperm__
41. The resting membrane potential is.....mv. _____70mV
42. Fusion is a __Active___process that is activated by fusogenic proteins.
43. Axoneme s formed by. _____Microtubules__
44. Eukaryotic genes are contained within a complex of DNA and protein called__Chromatin____
45. The negative enhancers are called___Silencers___
46. The ability of embryonic cells to change their fates is called.....___Regulation___
47. ____Pax6___transcription factor found in mammalian eye, nervous system and pancreas.
48. E- Cadherin is expressed on all early mammalian embryonic cells.
49. Annelids, molluscs and flatworms belong to..... ___ Lophotrochozoa
50. Potency___ of cell decrease if cell is committed.
51. Fertilization in Rana Pipiens usually lasts for about __3 (three)_ years.
52. An uncommitted cell can be described as ___Naive___
53. Lying immediately beneath the plasma membrane of the egg is a thin ___Shell__
54. Acroome can be considered as modified secretory_____ **vesicles**
55. Most marine organisms release their ___gametes___ into the environment
56. The fast block method is used to prevent_____Polyspermy__
57. Autonomous specification was first demonstrated in_____1887
58. In mouse ZP3 is attached to_____zona pellucid
59. TAFs used in_____transcription factor activity
60. in slow block method which is cortical granule protein_____ hyalin and peroxidase
61. Egg shell formed _____weight of egg._____9 – 12%
62. potency is __intrinsic__ property
63. cytoplasm that contains many nuclei called _____ syncytium
64. Ectoderm forms..... __ epidermis, nervous system, and pigment cells.
65. according to dutch sperm are ...___ parasitic animals (spermatozoa)
66. Acrosome & nucleus constitute the head of the sperm.
67. cortical granules are _____membrane bound structures
68. Acrosomal reaction has 2 (two) component
69. Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules
70. Ca & bicarbonate ions maybe critical in activating cAMP
71. Peroxidase enzyme hardens fertilization envelope
72. The Fast block to polyspermy is achieved by changing the electric potential
73. P-cadherin stands for _____Placental Cadherin
74. Most transcription factor can bind to specific DNA sequences
75. MITF protein necessary for production of pigments cell and their pigments.
76. Outside of plasma membrane is the vitelline envelope.
77. Mouse __**Fertilin**___ is localized to posterior plasma membrane of the sperm **head**
78. Finally the fourth cortical granules protein ___hyaline___ forms a coating around egg.

Join VU BIO ANIMALS

79. The cleavage stages are called. Blastomere
80. Enhancers are..... DNA Sequences
81. The mitotic spindle and contractile ring are perpendicular to each other
82. J.Whittaker stained blastomeres for the presence of the enzyme acetyl cholinesterase
83. Protocadherins are involved in separating notochord from mesoderm in Xenopus gastrulation
84. Pax6 factor needed for mammalian eye
85. Which stands comparative embryology is First
86. Cell type specification precedes any large-scale embryonic cell migration
87. Series of extremely rapid divisions in zygote is Cleavage
88. In protosome, mouth is formed first than Deuterostomes: forms.
89. Northern blots, in situ hybridization, & the PCR can show which cells are transcribing particular genes.
90. Coding regions of DNA are called Exon
91. Na and K Are important in fast block polyspermy.
92. The force for sperm propulsion is provided by dynein
93. Spallenazani demonstrated that filtered toad semen devoid of sperm would not fertilize egg.
94. Mesoderm generates the KIDNEY.
95. Sperms are attracted towards egg of their species by CHEMOTAXIS.
96. Mathematical modeling seeks to desire developmental phenomenon in terms of EQUATIONS.
97. A region that is not translated in proteins is called UNTRANSLATED region.
98. Invariant cleavage produce the SAME lineages in embryo
99. Germ layers are formed as a result of GASTRULATION
100. RNA polymerase binds with PROMOTOR in start of transcription
101. The region of globular actin molecule lies between nuclear and acrosomal vesicle in SEA URCHIN
102. Which technique is used in situ hybridization Northern blots
103. Conditional specification involves interaction with RELATIVE POSITIONS cells
104. Particular proteins or carbohydrate on the sperm surface LOST during capacitation
105. LOSS of hydrogen ion causes pH to rise.
106. Peroxidase enzyme hardens the fertilization envelope by crosslinking TYROSINE residues
107. FERTILIZATION transmits genes from parents to offspring.
108. Regulation of gene expression accomplished by SELECTIVE MESSENGER RNA TRANSLATION
109. Development contact genotype and PHENOTYPE
110. Nucleosome is composed of OCTAMER of histone protein.

1. Knowledge of gene activity in humans can be obtained by candidate GENE MAPPING & POSITIONAL CLONING
2. Anatomical abnormalities are caused by MUTANT GENES
3. All multicellular organisms arise by a slow process of progressive change called DEVELOPMNET
4. Animals development is the division of embryo into three layers Ectoderm, Mesoderm and ENDODERM
5. The ions which are responsible for increasing granule reactions Ca (Calcium)
6. The zona pellucida is a fglycoprotein matrix, which is synthesized and secreted by the growing oocyte
7. In several other species, certain regions of the membrane are specialized for sperm recognition & fusion
8. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development
9. Activation of egg metabolism to start development
10. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates
11. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in Acetabularia.
12. Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly
13. Bacteria are able to transmit genes from one individual to another by means of sex pili

10. +ve and – ve mating take place *Chlamydomonas*

Join VU BIO ANIMALS